HVAC System Optimization Report

Ottawa Commercial Building Case Study

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HVAC Optimization Project

Realistic TMYx Weather • Part-Load Curves • Monte Carlo Analysis

Executive Summary

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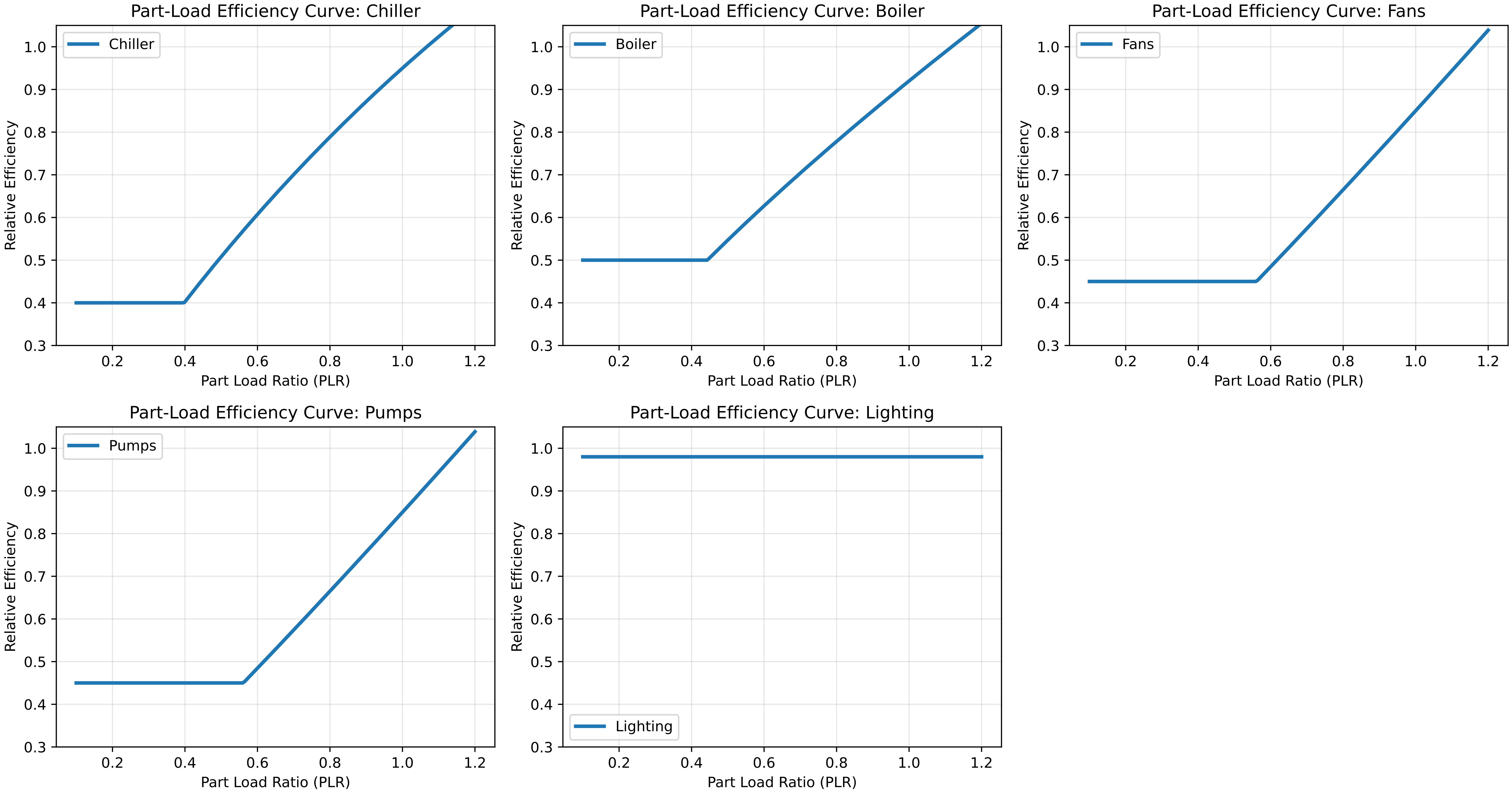
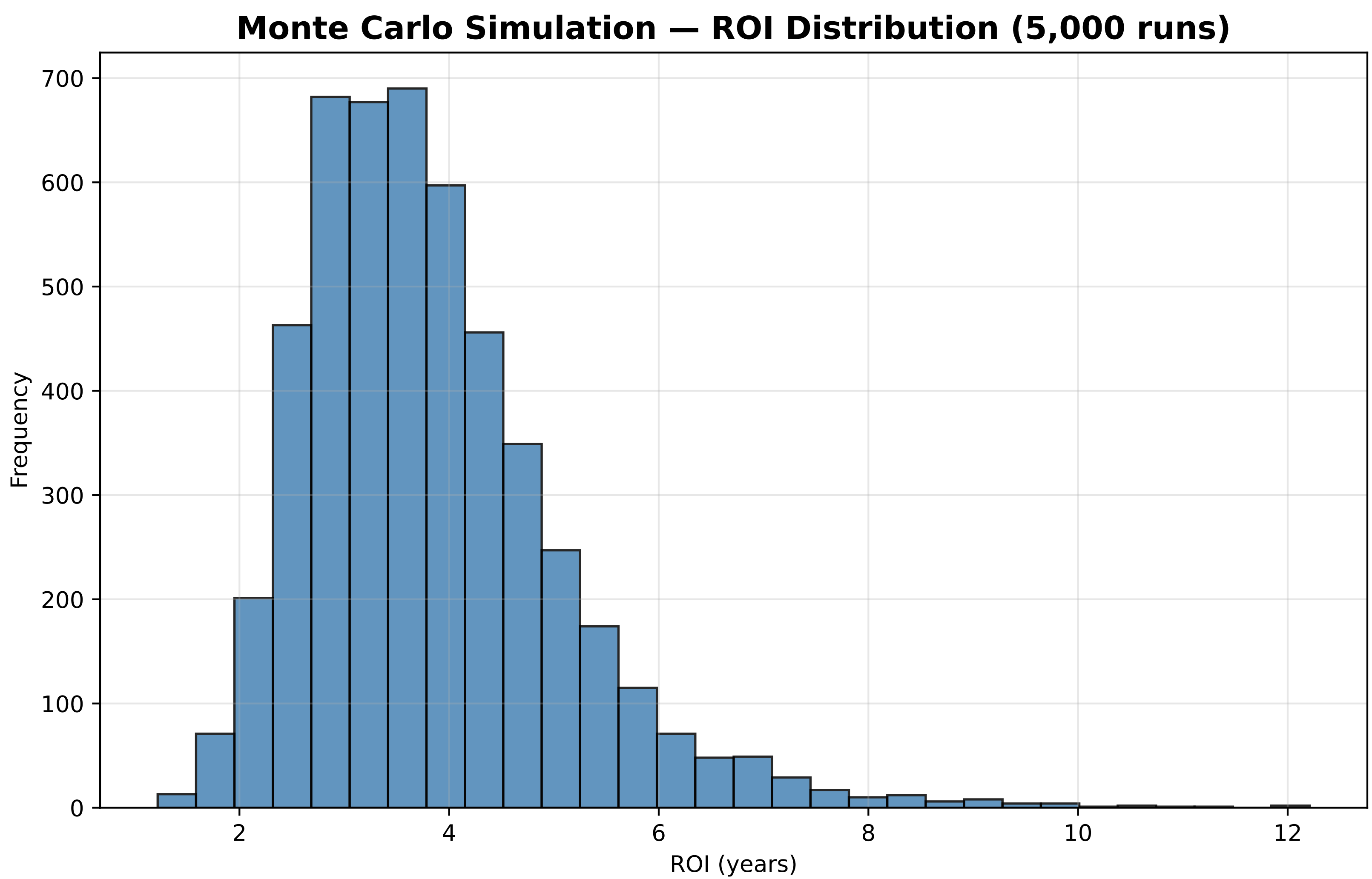
Baseline Performance
    Annual Energy Consumption : 636,413 kWh
    Annual Operating Cost      : $76,369.55

Best Scenario – Combined Measures
    Annual Energy              : 520,613 kWh
    Energy Reduction           : 115,800 kWh (18.2%)
    Cost Savings               : $13,896.02
    Payback Period             : 3.6 years
    CO₂ Reduction              : 57.9 tonnes

Parametric Optimization (SciPy)
    Additional Annual Savings  : $-8,969
    Minimum Annual Cost Found  : $53,504.14

Monte Carlo Analysis (5,000 runs)
    Mean ROI                   : 3.8 years
    95% Confidence Interval    : 2.0 – 6.8 years
    Probability of ROI < 8 yrs : 99.0%
```


Part-Load Efficiency Curve: Chiller
Chiller
Part-Load Efficiency Curve: Boiler
Boiler
Part-Load Efficiency Curve: Fans
Fans
Part-Load Efficiency Curve: Pumps
Pumps
Part-Load Efficiency Curve: Lighting
Lighting
Relative Efficiency
Part Load Ratio (PLR)
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0.2
0.4
0.6
0.8
1.0
1.2


Monte Carlo Simulation — ROI Distribution (5,000 runs)
Frequency
ROI (years)
0
100
200
300
400
500
600
700
2
4
6
8
10
12

Optimal Configuration from Parametric Optimization

Recommended Upgraded Component Ratings:

```
    Chiller      :  22.5 kW   ( 25.0% reduction)
    Boiler       :  12.0 kW   ( 20.0% reduction)
    Fans         :   3.2 kW   ( 36.0% reduction)
    Pumps        :   3.5 kW   ( 30.0% reduction)
    Lighting     :   3.2 kW   ( 60.0% reduction)
```